Definition 2.34. Let A be a definable subset of VF^n. The RV-*boundary* of A, denoted by $\partial_{\mathrm{RV}}A$, is the definable subset of $\mathrm{rv}(A)$ such that $t \in \partial_{\mathrm{RV}}A$ if and only if $t^\sharp \cap A$ is a proper nonempty subset of $t^\sharp$. The definable set $\mathrm{rv}(A) \smallsetminus \partial_{\mathrm{RV}}A$, denoted by $\mathrm{int}_{\mathrm{RV}}(A)$, is called the RV-*interior* of A.

Resolved referee R032 [Typo] – Definability of the RV-boundary.
Original source line 945. Insert 'definable' before 'subset of VF^n'; without that assumption the boundary and interior need not be definable.

Author resolution. The author approves explicitly assuming that A is definable, so that its RV-boundary and RV-interior have the stated definability.

Obviously, $A \subseteq \mathrm{VF}^n$ is an RV-pullback if and only if $\partial_{\mathrm{RV}}A$ is empty.

Lemma 2.35. *Let A be a definable subset of VF^n. Then $\dim_{\mathrm{RV}}(\partial_{\mathrm{RV}}A) < n$.*

Proof. For $n = 0$, the boundary is empty and has dimension $-\infty$. For $n = 1$, take a finite Boolean disc decomposition of A. A disc containing 0 is an RV-pullback, whereas a disc avoiding 0 lies in one RV-disc. A partially met RV-fiber therefore contains a member of this finite disc family that avoids 0, so there are only finitely many such fibers.

For the inductive step, fix $i \in [n]$ and put $X_i = \mathrm{pr}_{\tilde{i}}(A)$ and $T_a = \partial_{\mathrm{RV}}A_a$ for $a \in X_i$. Each T_a is finite. For each $t \in T_a$, Corollary 2.21, over the parameters (a, t), gives a definable point in $t^\sharp$. By compactness there are finitely many definable partial choice functions covering all such pairs (a, t); order this list and use the first available choice to obtain a definable function $c(a, t) \in t^\sharp$. Set $f_i(a) = \{c(a, t) : t \in T_a\}$ if $T_a \neq \varnothing$, and $f_i(a) = \{0\}$ otherwise. Thus every fiber is finite and contains exactly one representative of each boundary tag when that boundary is nonempty; in particular $\mathrm{rv} \upharpoonright f_i(a)$ is injective.

First split X_i by the pattern of its zero coordinates, and apply Corollary 2.30 on the remaining coordinates of each piece. This gives a definable finite partition $(A_{ij})_j$ of X_i such that, whenever $t^\sharp \subseteq A_{ij}$, the finite set $\mathrm{rv}(f_i(a))$ is independent of $a \in t^\sharp$. The zero-pattern split allows t to have ∞ coordinates: after deleting them, $t^\sharp$ is an open polydisc in the remaining coordinates. Write $f_{ij} = f_i \upharpoonright A_{ij}$ and let $C_{ij}(t)$ be this common finite set for $t \in \mathrm{int}_{\mathrm{RV}}(A_{ij})$.

Regard each C_{ij} as its graph in RV^n_∞, retaining the input tags and placing its output tag in coordinate i. Put

$$C = \bigcup_{i,j} C_{ij}, \qquad B = \bigcup_{i,j} \{s \in \mathrm{rv}(A) : \mathrm{pr}_{\tilde{i}}(s) \in \partial_{\mathrm{RV}}A_{ij}\}.$$

Each graph C_{ij} is finite over its $n-1$ input coordinates, so $\dim_{\mathrm{RV}}(C) < n$. By induction, $\dim_{\mathrm{RV}}(\partial_{\mathrm{RV}}A_{ij}) < n-1$; with only one remaining coordinate, this gives $\dim_{\mathrm{RV}}(B) < n$. Finally, for a partially met RV-polydisc, start in A and change coordinates one at a time to reach a point outside A. The first exit identifies a boundary tag of a unary section. Its other tags either lie in some $\partial_{\mathrm{RV}}A_{ij}$, placing the full tuple in B, or in $\mathrm{int}_{\mathrm{RV}}(A_{ij})$, placing it in C. Hence $\partial_{\mathrm{RV}}A \subseteq B \cup C$, as required. □

Resolved referee R033 [Gap] – RV-boundary proof needs uniform branches.
Original source line 962. When constructing f_{ij}, ensure that each fiber contains exactly one representative of each boundary RV-disc; injectivity of rv on these finite fibers is a hypothesis of Corollary 2.30. The sentence about constancy should mean that the finite set $\mathrm{rv}(f_{ij}(a))$ is independent of $a \in t^\sharp$, not that rv is constant on the union of all those finite sets. The argument also depends on the preceding corollary after its parameter and multidimensional gaps are repaired. A shorter alternative is the human referee's generic-lift argument: over a model containing the parameters, a tuple of full RV-dimension has a lift realizing one complete type, so its lift cannot meet A partially; state the dimension/type fact precisely if using this route.

Author resolution. The author approves choosing exactly one VF representative of each boundary RV tag and stating constancy of the finite RV-image set along each source RV fiber. The proof now splits zero-coordinate faces before applying Corollary 2.30, retains all input tags in the graphs, and gives the zero-dimensional base case and the coordinate-chain coverage argument.

Let $f : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ be a definable function.

Lemma 2.36 ([17, Lemma 8.7])**.** *There is a definable closed set $A \subseteq \mathrm{VF}^n$ with $\dim_{\mathrm{VF}}(A) < n$ such that $f \restriction (\mathrm{VF}^n \smallsetminus A)$ is continuous.*

Per the usual terminology, this lemma says that f is continuous *almost everywhere*.

Definition 2.37. For any $a \in \mathrm{VF}^n$, we say that f is *differentiable at a* if there is a linear map $\lambda : \mathrm{VF}^n \longrightarrow \mathrm{VF}^m$ (of VF-vector spaces) such that, for any $\epsilon \in \Gamma$, if $b \in \mathrm{VF}^n$ and $\mathrm{val}(b)$ is sufficiently large then

$$\mathrm{val}(f(a+b) - f(a) - \lambda(b)) - \mathrm{val}(b) > \epsilon.$$

It is straightforward to check that if such a linear function λ (represented by a matrix with entries in VF) exists then it is unique and hence is called the *derivative of f at a*.

Write $f = (f_1, \ldots, f_m)$. For $a = (a_i, a_{\tilde{i}}) \in \mathrm{VF}^n$, if the derivative of the function $f_j \restriction (\mathrm{VF} \times a_{\tilde{i}})$ at a_i exists then we call it the *ijth partial derivative of f at a*.

Referee R034 [Clarification; revised] – Valuation of a tuple and the zero increment.
Original source line 980. Define $\mathrm{val}(b) = \min_i \mathrm{val}(b_i)$ for a nonzero tuple, with $\mathrm{val}(0) = \infty$, and require $b \neq 0$ in the difference quotient. Otherwise the display includes the undefined expression $\infty - \infty$ at $b = 0$. Later, where valuations of tuples are taken coordinatewise (Terminology 2.39), distinguish that use from this minimum convention. The input-first indexing of partial derivatives is legitimate, but please keep the corresponding matrix convention explicit.

We differentiate functions between arbitrary definable sets by "forgetting" the RV-coordinates. More precisely, let

$$f : \mathrm{VF}^n \times \mathrm{RV}^m \longrightarrow \mathrm{VF}^{n'} \times \mathrm{RV}^{m'}$$

be a definable function. By dimension theory, for every $t \in \mathrm{RV}^m$ there is an $s \in \mathrm{RV}^{m'}$ such that $\dim_{\mathrm{VF}}(\mathrm{dom}(f_{(t,s)})) = n$ and hence $\mathrm{dom}(f_{(t,s)})$ has an open subset. For such an $s \in \mathrm{RV}^{m'}$ and each a contained in an open subset of $\mathrm{dom}(f_{(t,s)})$, we define the partial derivatives of f at (a,t) to be those of $f_{(t,s)}$ at a. If $n = n'$ and all the partial derivatives exist at a point (a,t) then the *Jacobian of f at (a,t)* is defined in the usual way (that is, the determinant of the Jacobian matrix) and is denoted by $\mathrm{Jcb}_{\mathrm{VF}} f(a,t)$.

Lemma 2.38. *The Jacobian of f exists and is a continuous function almost everywhere.*

Proof. This is immediate by [17, Corollary 9.9]. □

Referee R035 [Gap] – Equal VF dimensions and generic RV constancy.
Original source line 993. State $n = n'$ in Lemma 2.38; otherwise the determinant called the Jacobian has not been defined. Moreover, the preceding existence of one full-dimensional fiber for each input t does not show that such fibers cover almost every input point. Use the standard generic local constancy of a definable map from VF^n to $\mathrm{RV}^{m'}$ first, and then apply the cited differentiability theorem on its open fibers. This provides the missing almost-everywhere coverage.